

A Speech-Based Conversational Recipe Recommendation Agent

Project Conversational Agents - Lecture 1

Overview

Project Conversational Agents

- **Conversational agents**
- Assignment introduction
- Project organization
- Dialogflow

What is a conversational agent?



Joe Weizenbaum's ELIZA (1965): Scripted templates

Human: Men are all alike.

ELISA: In what way?

Human: They're always bugging us about something.

ELISA: Can you think of a special example?

Human: Well, my boyfriend made me come here.

ELISA: Your boyfriend made you come here?

Human: He says I'm depressed much of the time.

ELISA: I'm sorry to hear that you are depressed much of the time.

Human: It's true. I'm unhappy.

ELISA: Do you think coming here will help you not to be unhappy?

Human: I need some help, that much seems certain.

○ ○ ○

Eliza (1966)

```
> Hello, I am Eliza.  
* Hey!  
> Don't you ever say Hello?  
* Hello!  
> How are you today.. What would you like to discuss?  
* I'm depressed.  
> Do you enjoy being depressed?  
* No, I don't.  
> Why no?  
* Because I want to be happy.  
> Why do you want to be happy?  
* Because I'm tired od being depressed.  
> That is quite interesting.  
* I'm not sure█
```

IBM Watson (2011): Question-Answering in Jeopardy!



Microsoft's Tay (2016): Chatbot

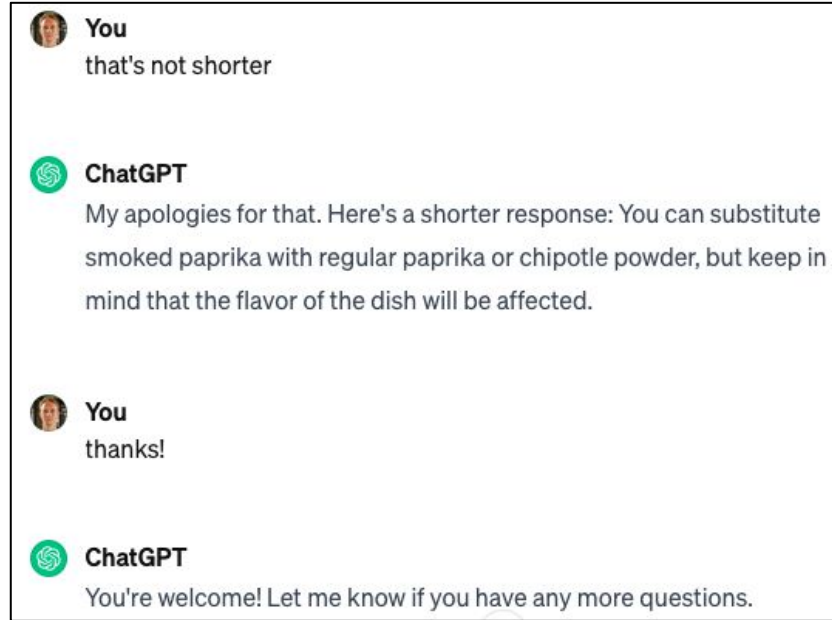


Tay (Think-About-You) began to post inflammatory and offensive tweets <24h.

ChatGPT (2022): Natural language generation



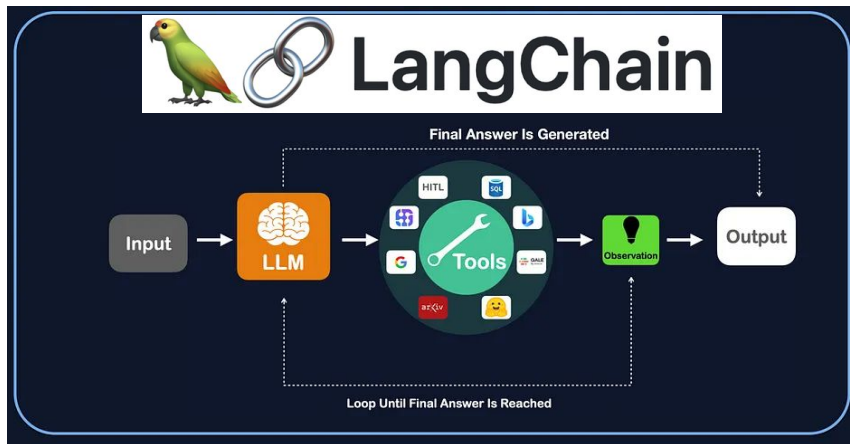
Large Language Model
(LLM)



How's your experience?

Next-Gen LLMs since 2023

Langchain / AutoGen



LLMs + Tools + Memory + Planning + Action + ... → SOTA Conversational Agent Systems

What is a conversation?

A: Hi, I'm Alma. I'm here to answer questions you have about your Medicare insurance. How can I help you?

U: I have a question about diabetes screenings

A: Okay, what's your question?

U: am I covered for them?

A: Diabetes screenings are covered at 100% of the approved amount, so there is no copay or deductible if you see doctors who take assignment.

U: what do you mean by take assignment?

A: Oh. Assignment means that your doctor, provider or supplier agrees to accept the Medicare-approved amount as full payment.

U: ok thanks

A: You're welcome! Is there anything else I can do for you?

Understanding a conversation

A conversational agent understands what has been said, for example,

User: I have a question about diabetes screenings.

The agent should understand:

- The user's intention is to ***request*** an information
- The requested information is about ***diabetes screenings***

Managing a conversation

A conversational agent remembers what has been said and can perform social actions for managing the conversation itself, such as:

- Handling misunderstandings:
“What did you say?,” “none of your business,” “okay, never mind”
- Explaining itself
- Handling repeats
- ...

Talking about talk: Operations on prior utterances in the interaction. The social actions they perform constitute a major part of what makes a conversation a conversation.



Conversational agents are computer programs that can simulate a conversation with a human using speech or text (natural language).

Extended multi-turn conversations (> 10 turns)

Some challenges:

- Conversation can go in many directions.
- Maintain coherence throughout conversation.

Solution 1:

Let the user do the talking

Solution 2:

Let the agent guide the conversation

Solution 3:

Utilize common conversation structure

Solution 1: Joe Weizenbaum's ELIZA (1965)

Human: Men are all alike.

ELISA: In what way?

Human: They're always bugging us about something.

ELISA: Can you think of a special example?

Human: Well, my boyfriend made me come here.

ELISA: Your boyfriend made you come here?

Human: He says I'm depressed much of the time.

ELISA: I'm sorry to hear that you are depressed much of the time.

Human: It's true. I'm unhappy.

ELISA: Do you think coming here will help you not to be unhappy?

Human: I need some help, that much seems certain.



The screenshot shows a terminal window titled "Eliza (1966)". The conversation is as follows:

```
> Hello, I am Eliza.
* Hey!
> Don't you ever say Hello?
* Hello!
> How are you today.. What would you like to discuss?
* I'm depressed.
> Do you enjoy being depressed?
* No, I don't.
> Why no?
* Because I want to be happy.
> Why do you want to be happy?
* Because I'm tired od being depressed.
> That is quite interesting.
* I'm not sure
```

Solution 2 example:

Conversation mostly guided by agent

Defined/limited options for user to respond

- 1 *Which team are you rooting for?*
- 2 • *Yankees.*
- 3 • *Red Sox.* ⇐
- 4 *Really? But they aren't so great at winning the World Series!*
- 5 • *I bet you are a Yankees fan.* ⇐
- 6 • *Ah, but who cares? They play great ball!*
- 7 *No, I'm just joking with you.*
- 8 • *Oh.* ⇐
- 9 • *That's too bad, it would be more fun if you were!*
- 10 *Did you catch Thursday's game?*
- 11 • *Yes.* ⇐
- 12 • *No.*
- 13 • *I don't want to talk about baseball anymore.*
- 14 *What did you think of it?*
- 15 • *Awesome game!* ⇐
- 16 • *It was boring.*
- 17 • *We really needed that win.*
- 18 *Yeah, it was very intense. Great game.*
- 19 • *Yeah.* ⇐
- 20 *I wonder how the Red Sox are doing in the standings. Should I check?*
- 21 • *Yes.*
- 22 • *No.* ⇐
- 23 • *I don't want to talk about baseball anymore.*

Solution 3 example

Utilizing structure of conversational activities / genres

Bakery:

B. "Who's next?"

C. "Could I have a cut wholegrain, please?"

B. "Coming right up!"

B. "Anything else?"

C. "Do you have tompouce?"

B. "All sold out, I'm afraid"

C. "Oh, too bad. That is all then."

B. "3 euro's, please."

C. "Goodbye!"

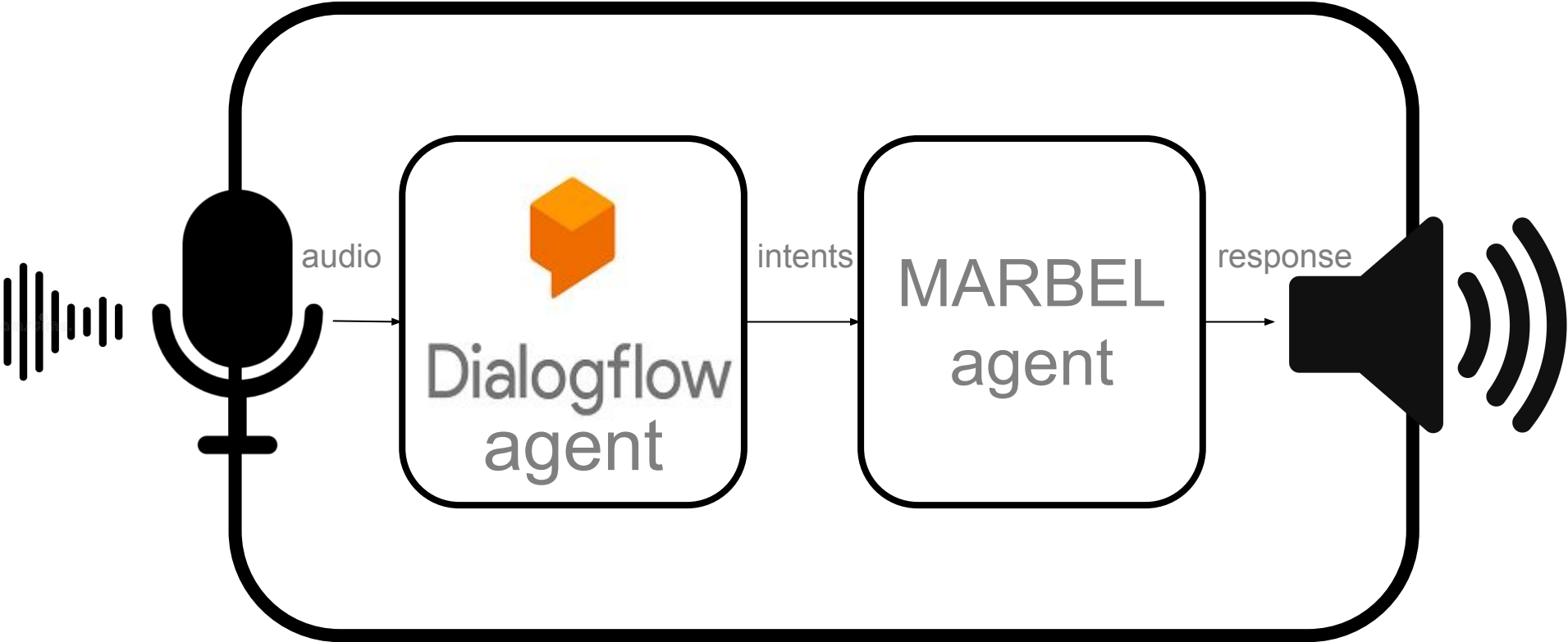
B. "Have a nice day!"



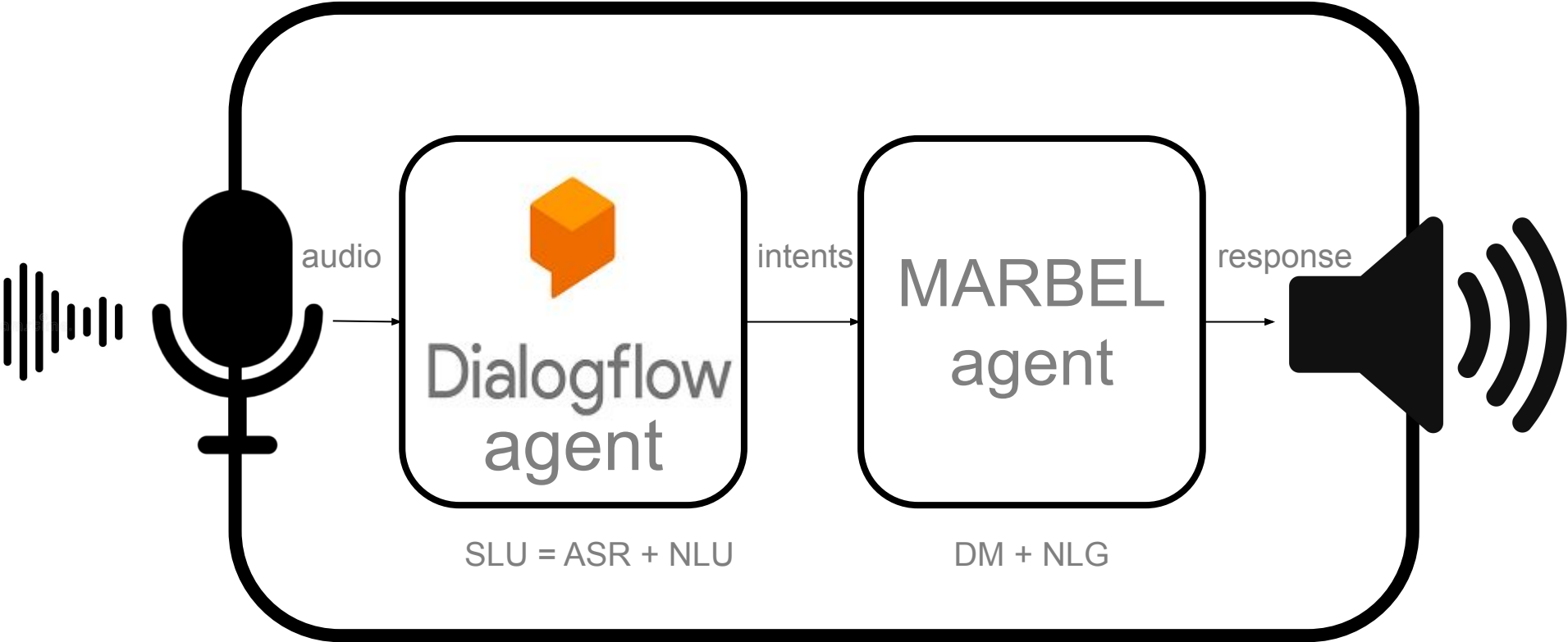
A peek under the hood



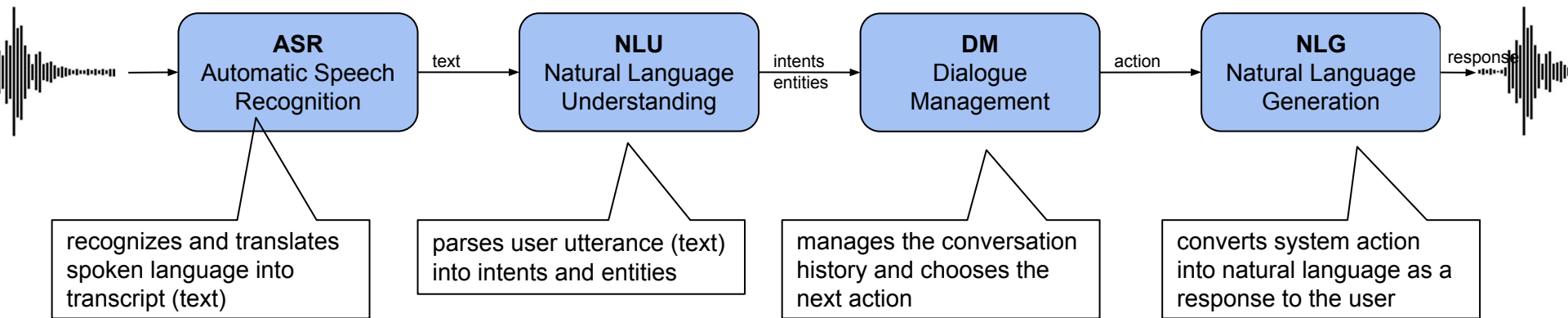
Our conversational agent is built out of other agents



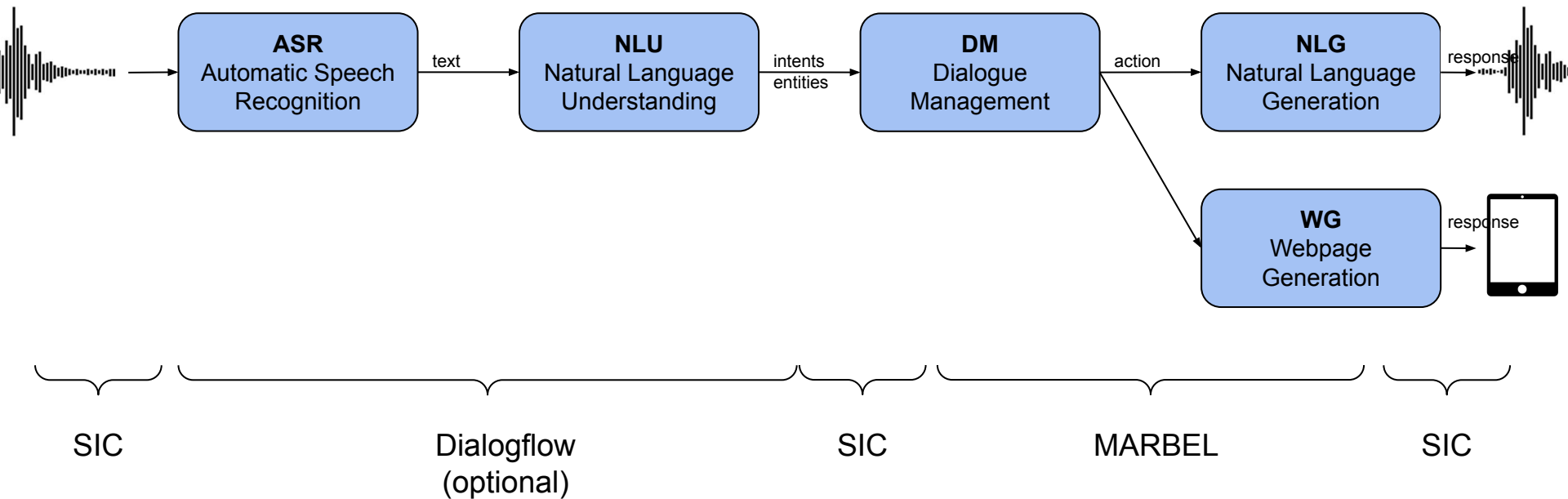
Our conversational agent is built out of other agents



Pipeline Architecture



Pipeline Architecture: In this project



Overview

Project Conversational Agents

- Conversational agents
- **Assignment introduction**
- Project organization
- Dialogflow

Task-based conversations: Recipe recommendation



A search task: find recipe in a database that matches user preferences

- Prolog database with about 900 recipes made available to you

Traditional internet search approach: provide Search Engine Results Pages (SERPs) in response to a user query

What is a conversational approach?

Task-based conversations: Recipe recommendation



Closed-domain / task-based

- Knowledge about cooking domain
- Database of recipes

Task-based conversations: Recipe recommendation



Closed-domain / task-based

- Knowledge about cooking domain
- Database of recipes

Model-based (conversational patterns)

Task-based conversations: Recipe recommendation



Closed-domain / task-based

- Knowledge about cooking domain
- Database of recipes

Model-based (conversational patterns)

Extended length

- Utilizing structure of recipe recommendation genre and conversational activities
-

Task-based conversations: Recipe recommendation



Closed-domain / task-based

- Knowledge about cooking domain
- Database of recipes

Model-based (conversational patterns)

Extended length

- Utilizing structure of recipe recommendation genre and conversational activities

Audio + visual interface

What is a good task-based conversational agent?

(Or: when do you have a bad experience with a conversational agent?)

- **Conversational:** Can converse coherently, provides **sensible and specific** responses
- **Effective:** Can provide the **assistance** the user need
- **Efficient:** Can be used **intuitively**, or communicate about its capabilities to the user
- **User experience:** Can **engage** the user
- **Robust:** Does not break down easily, can recover from **misunderstandings**

Demo

Try it out!

- [Basic Agent with Dialogflow](#)
- [Basic Agent with ASR+NLU](#)

Overview

Project Conversational Agents

- Conversational agents
- Assignment introduction
- **Project organization**
- Dialogflow

Introducing the teachers and TAs



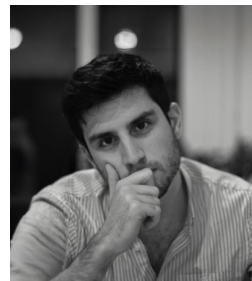
Jiahuan



Koen



Lazar
(Super TA)



Jacopo
(JT/TA)



Ahmad
(TA)



Alp
(TA)



Enea
(TA)



Hanna
(TA)



Mateusz
(TA)



Nikolay
(TA)



Teodora
(TA)



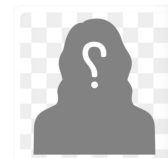
Vanshita
(TA)



Prakhar
(TA)



Snigdha
(TA)



Raina
(TA)

The place to be:

PCA 2026 confluence pages

<https://socialrobotics.atlassian.net/wiki/spaces/PCA21/pages/3117779057/Home+-+PCA+2026+Overview>

Solution 1: Your MARBEL Agent

- Dialogflow + MARBEL Agent

▼ Your MARBEL Agent

- Getting Started with your MARBEL Agent
- Greet, and Self-Identify
- Request a Recommendation
- Select Recipes by Name
- Filtering by Inclusion
- Connecting Your Entire Pipeline
- Handle Unexpected Intents
- Confirmation and Closing
- Remove a Filter and See Recipes on Demand
- Visuals and Visual Requirements



**Some capabilities
will require MORE
EFFORT than
others!!**

For example, some of the filtering capabilities require a significant amount of work!

**Separate tasks can be done in parallel!
You want to split the work between team members!**

Solution 2: Your Intent and Slot Classifier

- NLU model $\Rightarrow$ Your Intent and Slot Classifier
- ASR+NLU+MARBEL Agent

▼ Your Intent and Slot Classifier

- Ontology
- Fitting Encoders
- Our Dataset
- Your NLU Model
- Training Your Model
- Model Evaluation
- Model Improvement

```
..... Talk now Round 1
[NLUComponent 192.168.2.12]: INFO: Running inference on example text: Can you teach me how to cook Italian pasta?
[NLUComponent 192.168.2.12]: INFO: Inference result: addFilter, 0.861081863747406, ['O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-cuisine' 'I-recipe' 'O' 'O' 'O' 'O' 'O' 'O'], [0.923848734664917, 0.9994171857833862, 0.999616265296936, 0.9987136125564575, 0.9986129999160767, 0.99956947565087874, 0.9992814660072327, 0.3889587092399597, 0.5767447352409363, 0.9918792843818665, 0.998571515083313, 0.9999287128448486, 0.9999319314956665, 0.9999321699742456, 0.999931812286377, 0.9999336004257202], 0.05375218391418457
Transcript: Can you teach me how to cook Italian pasta?
Intent: addFilter
0.861081863747406
Slots:
['O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-cuisine' 'I-recipe' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O']
.....
..... Talk now Round 2
[NLUComponent 192.168.2.12]: INFO: Running inference on example text: Can you teach me how to cook Chinese noodles?
[NLUComponent 192.168.2.12]: INFO: Inference result: addFilter, 0.9534512162208857, ['O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-cuisine' 'I-recipe' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O'], [0.9553836093711853, 0.9994514584541321, 0.999669875812207, 0.9998548491477966, 0.9998678429603577, 0.9995872378349384, 0.9993963241577148, 0.4168327911376953, 0.38481736183166504, 0.971748245819092, 0.9998049084554749, 0.9999276399612427, 0.9999178647994995, 0.9999257326126899, 0.9999299849377441, 0.9999288328541382], 0.05116222428894843
Transcript: Can you teach me how to cook Chinese noodles?
Intent: addFilter
0.9534512162208857
Slots:
['O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-cuisine' 'I-recipe' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O']
.....
..... Talk now Round 3
[NLUComponent 192.168.2.12]: INFO: Running inference on example text: Can you give me a touch broger?
[NLUComponent 192.168.2.12]: INFO: Inference result: addFilter, 0.9291972517967224, ['O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-tag' 'I-ingredient' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O'], [0.9541722536087836, 0.9994243383407593, 0.999588668346405, 0.999836381912231, 0.9984168661697388, 0.9404817925262451, 0.3699498427970886, 0.5918263792991638, 0.9950283895568848, 0.99295353894653, 0.999926293557739, 0.9999438179595947, 0.9999510049819946, 0.9999521970748981, 0.9999531507492065, 0.9999512434005737], 0.05537706653076172
Transcript: Can you give me a touch broger?
Intent: addFilter
0.9291972517967224
Slots:
['O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-tag' 'I-ingredient' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O']
.....
..... Talk now Round 4
[NLUComponent 192.168.2.12]: INFO: Running inference on example text: I would like a Coca-Cola and coffae.
[NLUComponent 192.168.2.12]: INFO: Inference result: addFilter, 0.6093176007270813, ['O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-tag' 'I-ingredient' 'I-recipe' 'O' 'O' 'B-ingredient' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O'], [0.9653940280885664, 0.9995625615119934, 0.9997429251670837, 0.9997512698173523, 0.42384458273513794, 0.36839917302131653, 0.3322422808374603, 0.97640776634241631, 0.3161783944286238, 0.9827704429262465, 0.9852676391601562, 0.9999275207519531, 0.9999327659606934, 0.9999293808912964, 0.9999265670776307, 0.9999277591705322], 0.04249691963195801
Transcript: I would like a Coca-Cola and coffae.
Intent: addFilter
0.6093176007270813
Slots:
['O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'O' 'B-tag' 'I-ingredient' 'I-recipe' 'O' 'O' 'B-ingredient' 'O' 'O' 'O' 'O' 'O' 'O' 'O']
```

Pilot User Study

Week 4

Report on your
findings



For assessment, check out:

<https://socialrobotics.atlassian.net/wiki/spaces/PCA21/pages/3117780447/Project+Deliverables>

What's New:

digital exam on Jan 19 2026

Onsite Exam

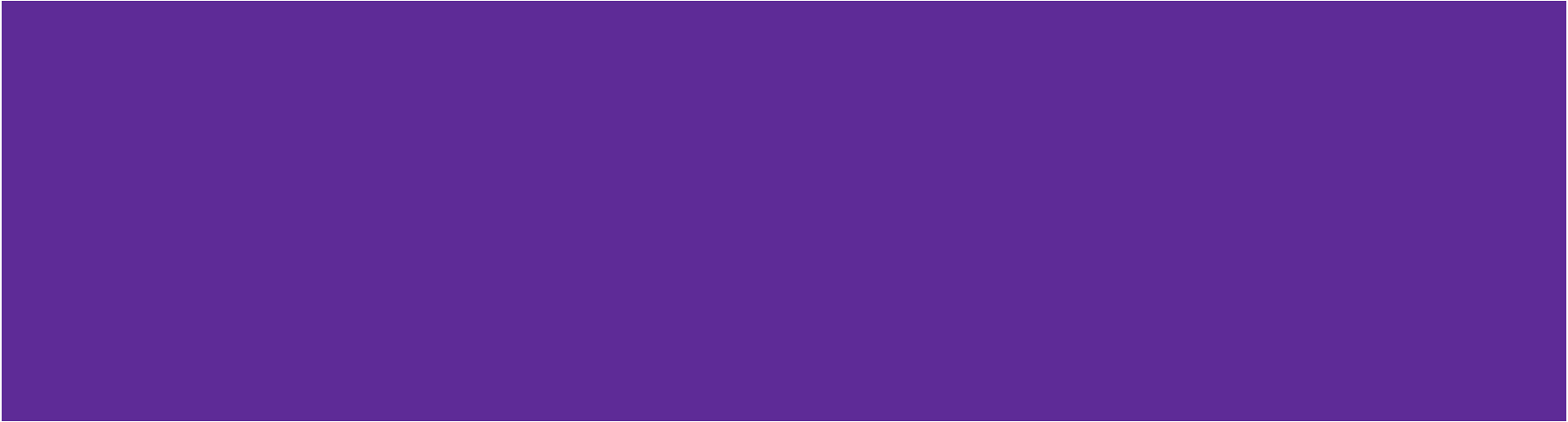
- January 19th 2026, from 15:30 to 16:30 (17:00 for extra time students)
 - More info about location to be announced on Canvas
- 30 Multiple-Choice Questions based on Preliminaries page (on Confluence) & Lectures 1, 2, and 3
- 15% of the final grade
 - No minimum passing grade

Overview

Project Conversational Agents

- Conversational agents
- Assignment introduction
- Project organization
- **Dialogflow**

Connecting your Dialogflow agent with your Dialog manager agent (MARBEL)





Create Agent



Docs

Dialogflow CX ^[new]



Support [↗](#)



Account



Logout

JeroenGroet

CREATE

DEFAULT LANGUAGE [?]

English – en ▾

Primary language for your agent. Other languages can be added later.

DEFAULT TIME ZONE

(GMT+1:00) Europe/Madrid ▾

Date and time requests are resolved using this timezone.

GOOGLE PROJECT

New GCP project will be automatically linked to the agent after saving

AGENT TYPE



Set as Mega Agent

Combine multiple Dialogflow agents (i.e. sub agents) into a single agent (i.e. [mega agent](#)).



Please, create at least one
agent to access the test
console

Dialogflow ES

Documentatie

Basisbeginselen

Introductievideo's

Dialogflow ES-console 

Snelstartgidsen

Alle snelstartgidsen

Instellen

Een agent ontwerpen

Fulfillment (uitvoeringscode)

Interacties met een integratie

Interacties met de API

Concepten

Alle concepten

Overzicht van Dialogflow-console

▸ Agents

▸ Intenties

▸ Entiteiten

▸ Contexten

▸ Evenementen

▸ Fulfillment (uitvoeringscode)

▸ Integraties

▸ Kwaliteit en controle van agents

API-interacties

Kenniscconnectors 

Training

Toegangsbeheer

Meerdere projecten gebruiken

Praktische tips

Handleidingen

Alle handleidingen

For the purpose of trying the Dialogflow API, you can use the **Project > Owner** role in steps below, which grants the service account full access to the project. For more information on roles specific to Dialogflow, see the [Dialogflow access control document](#).

About service account keys

Service accounts are associated with one or more public/private key pairs. When you create a new key pair, you download the private key. Your private key is used to generate credentials when calling the API. You are responsible for security of the private key and other management operations, such as key rotation.

Create a service account and download the private key file

Set up authentication:

1. In the Cloud Console, go to the **Create service account key** page.

[Go to the Create Service Account Key page](#)

2. From the **Service account** list, select **New service account**.

3. In the **Service account name** field, enter a name.

4. From the **Role** list, select **Project > Owner**.



Note: The **Role** field affects which resources your service account can access in your project. You can revoke these roles or grant additional roles later. In production environments, do not grant the Owner, Editor, or Viewer roles. For more information, see [Granting, changing, and revoking access to resources](#).

5. Click **Create**. A JSON file that contains your key downloads to your computer.

Use the service account key file in your environment

Provide authentication credentials to your application code by setting the environment variable `GOOGLE_APPLICATION_CREDENTIALS`. Replace `[PATH]` with the file path of the JSON file that contains your service account key. This variable only applies to your current shell session, so if you open a new session, set the variable again.

Inhoudsopgave

[Before you begin](#)[About the Google Cloud Console](#)[Create a project](#)[Enable billing](#)[Enable the API](#)[Set up authentication](#)[About service accounts](#)[About roles](#)[About service account keys](#)[Create a service account and download the private key file](#)[Use the service account key file in your environment](#)[Install and initialize the Cloud SDK](#)[Test the SDK and authentication](#)[Install the Dialogflow client library](#)

Google Cloud Platform

jeroengroet-teci

Search products and resources

APIs & Services

Credentials

+ CREATE CREDENTIALS

DELETE

Dashboard

Library

Credentials

OAuth consent screen

Domain verification

Page usage agreements

Create credentials to access

Remember to c

API keys

No API keys to display

OAuth 2.0 Client

No OAuth clients to displ

Service Account

No service accounts to display

Key

Client ID

Manage service accounts

Select from VU.NL

NEW PROJECT

Search projects and folders

RECENT ALL

Name	ID
✓ jeroengroet-teci ?	jeroengroet-teci
blablabot-wnfk ?	blablabot-wnfk
JeroenGroet ?	jeroengroet

Create a service account!

CANCEL OPEN

Create service account key

Service account

New service account

Service account name

cooking

Role

Owner

Service account ID

cooking@jeroengroet-teci.iam.gserviceaccount.com

Key type

Downloads a file that contains the private key. Store the file securely because this key cannot be recovered if lost.

☒ JSON

Recommended

☐ P12

For backward compatibility with code using the P12 format

Create

Cancel

jeroengroet-teci-388ea823bf56.json openen

U hebt gekozen om het volgende bestand te openen:

jeroengroet-teci-388ea823bf56.json

van het type: Document (2,3 KB)

vanaf: data:

Wat moet Firefox met dit bestand doen?

☐ Openen met Xcode (standaard)

☒ Bestand opslaan

☐ Dit vanaf nu automatisch doen voor dit type bestanden

Annuleren

OK

CLOSE

Download the JSON key file!

Intent matching

AKA: intent recognition, intent classification, intent detection, “what the user wants to do”

What is intent matching?

Intent matching is classifying user expressions or utterances into predefined intent labels.

$$f(\text{utterance}) \rightarrow \{ \text{Intent}_1, \text{Intent}_2, \dots, \text{Intent}_N \}$$

Intent classifiers are typically implemented as a *multi-class classifier* neural network.

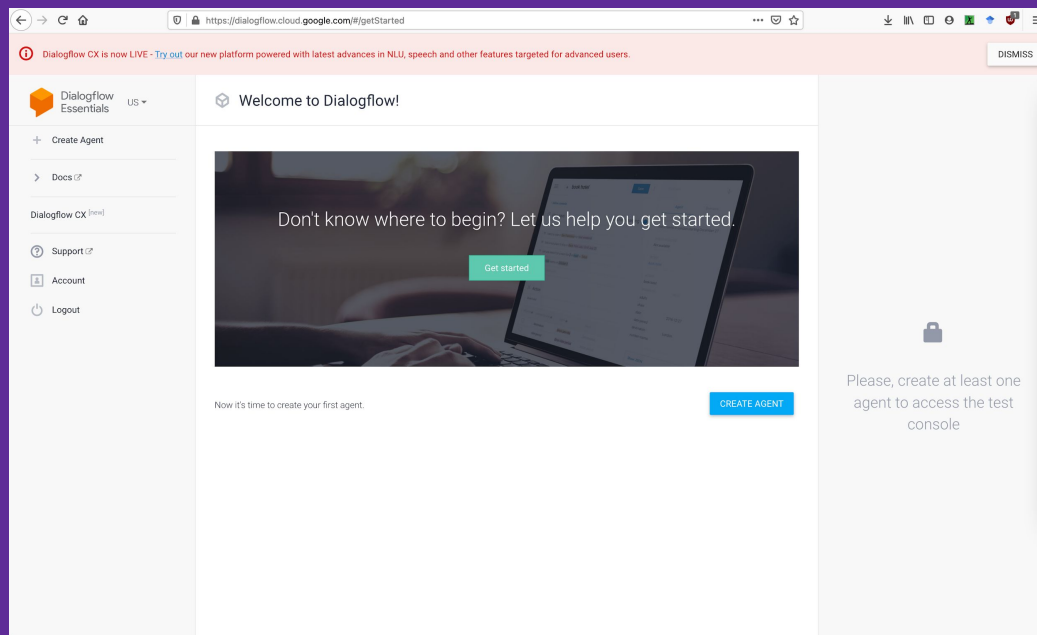
Examples:

$f(\text{"Hello"}) = \text{greeting.}$

$f(\text{"A recipe from Japan"}) = \text{addFilter. (a feature request)}$

Dialogflow
hides the
machine
learning
model.

You can treat it as a black box.



Intent matching in Google Dialogflow

Basic steps:

- Create an intent: a **label** (or name for your intent).

JeroenGroet ▾ ⚙

en

+

 Intents

+

 Entities

+

 Knowledge ^[beta] Fulfillment Integrations Training Validation History Analytics Prebuilt Agents Small Talk Docs Trial
Free

Upgrade

Dialogflow CX ^[new]

Search intents

 Default Fallback Intent Default Welcome Intent

No regular intents yet. [Create the first one.](#)

Intents are mappings between a user's queries and actions fulfilled by your software. [Read more here.](#)

Before you start, check out [Prebuilt Agents](#), a collection of agents developed by the Dialogflow team.

 Please use test console above to try a sentence.



JeroenGroet ▾ ⚙

en +



Intents +



Entities +

Knowledge ^[beta]

Fulfillment



Integrations



Training



Validation



History



Analytics



Prebuilt Agents



Small Talk

Docs [🔗](#)

Trial

Free

[Upgrade](#)Dialogflow CX ^[new]● Intent name

SAVE



Try it now



❗ Please use test console above to try a sentence.

Contexts ? ▾

Events ? ▾

Training phrases ? ^



Train the intent with what your users will say

Provide examples of how users will express their intent in natural language. Adding numerous phrases with different variations and parameters will improve the accuracy of intent matching. [Learn more](#)

[ADD TRAINING PHRASES](#)

Action and parameters ^



Extract the action and parameters

Parameters are specific values extracted from a user's request when entities are matched. The values captured by parameters can be used in fulfillment, or in building a response. If you mark parameters as required, Dialogflow will prompt the user if their values were not extracted from their initial request. [Learn more](#)

[ADD PARAMETERS AND ACTION](#)

Responses ? ^



Execute and respond to the user

Intent matching in Google Dialogflow

Basic steps:

- Create an intent: a **label** (or name for your intent).
- Add Training phrases: **example phrases** for what users might say.



JeroenGroet ▾ ⚙

en +

Intents +

Entities +

Knowledge ^[beta]

⚡ Fulfillment

🔄 Integrations

🎓 Training

✅ Validation

🕒 History

📊 Analytics

📁 Prebuilt Agents

💬 Small Talk

> Docs [🔗]

Trial

Free

[Upgrade](#)

Dialogflow CX ^[new]

• Greeting

SAVE



Contexts [?] ▾

Events [?] ▾

Training phrases [?]



”

” Good night

” Good evening

” Good afternoon

” Good morning

” Nice day

” Good day

” Greetings

” Hello

” Hi there

” Hi

Try it now



ⁱ Please use test console above to try a sentence.

Intent matching in Google Dialogflow

Basic steps:

- Create an intent: a **label** (or name for your intent).
- Add Training phrases: **example phrases** for what users might say.
- Fill the Action field: fill in the intent **label** *that the Dialog manager will use*.

JeroenGroet

en

Intents

Entities

Knowledge ^[beta]

Fulfillment

Integrations

Training

Validation

History

Analytics

Prebuilt Agents

Small Talk

> Docs

Trial
Free

Upgrade

Dialogflow CX ^[new]

Greeting

SAVE

Action and parameters

Greeting

REQUIRED

PARAMETER NAME

ENTITY

VALUE

IS LIST

☐

Enter name

Enter entity

Enter value

☐

+ New parameter

Responses



Execute and respond to the user

Respond to your users with a simple message, or build custom rich messages for the integrations you support. [Learn more](#)

[ADD RESPONSE](#)

Fulfillment

☒

Enable webhook call for this intent

☐

Enable webhook call for slot filling

Try it now



Please use test console above to try a sentence.

**The Dialog manager agent
receives the action value
from your Dialogflow agent
via the
Social Interaction Cloud.**

Intent matching in Google Dialogflow

Basic steps:

- Create an intent: a **label** (or name for your intent).
- Add Training phrases: **example phrases** for what users might say.
- Fill the Action field: fill in the intent **label** *that the Dialog manager will use*.
- SAVE!

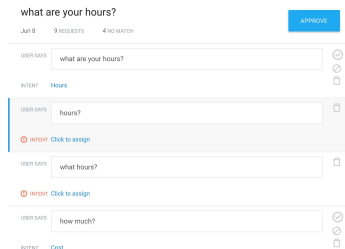
After saving, Dialogflow trains a model for intent classification for your intents.

Training phrases: dataset for training the classifier

Building a dataset for Intent Classification: Constructing a high-quality dataset involves data collection, annotation, labelling guidelines, diversity, and data augmentation techniques.

What does Google say:

- For each intent, you create **many training phrases**. When an end-user expression resembles one of these phrases, Dialogflow matches the intent.
- at least **10-20** (depending on complexity of intent, variety of end-user expressions)



Dialogflow uses two algorithms to match intents

Rule-based grammar matching

Accurate with few training phrase examples.

For matching, end-user expressions need to match almost completely verbatim with training phrases.

Machine Learning matching

Accurate *only* with numerous training phrase examples.

You don't have to define every possible example. For matching, Dialogflow's built-in machine learning expands on your training phrase examples list with other, similar phrases.

Intent detection confidence score and threshold

Dialogflow scores potential matches with an **intent detection confidence**, a value ranging from 0.0 (completely uncertain) to 1.0 (completely certain).

Rule-based grammar matching

A confidence score of 1 is indicative of rule-based matching.

Machine Learning matching

A confidence score < 1 is indicative of ML matching.

ML Classification Threshold

- If the highest scoring intent has a confidence score $\geq$ the threshold, it is returned as a match.
- If no intents meet the threshold, a fallback intent is matched.

ML Classification Threshold

General Languages **ML Settings** Export and Import Environments Speech Share Advanced

ML CLASSIFICATION THRESHOLD
Define the threshold value for the confidence score. If the returned value is less than the threshold value, then a fallback intent will be triggered, or if there is no fallback intents defined, no intent will be triggered.

0.3

AUTOMATIC SPELL CORRECTION
☐ Allow ML to correct spelling of query during request processing

AUTOMATIC TRAINING
☐ Disable automatic re-training the agent after every agent message.
Automatic training may slow UI responsiveness and is not recommended for production use. To enable automatic training, click the [API](#) or by clicking the "TRAIN" button below.

AGENT VALIDATION
☒ Automatically validate the agent when agent training is performed

TRAIN

Best
practice
for ASR.

**Default threshold
is quite low!**



Note: Before adjusting this value, you should try adding [negative examples](#).



Some intents
require **MANY
MORE TRAINING
EXAMPLES** than
others!!

*For example, the `addFilter` intent requires **many, many more** examples than a *greeting* intent.*

Ambiguity and Multi-Intent Interactions



Dialogflow

User queries are often ambiguous and can convey multiple intents simultaneously. Handling such ambiguity and multi-intent interactions can be challenging:

- **Ambiguity:** Users may phrase their queries in ways that can be interpreted differently. For example, “Book a flight from New York to Paris” could have both booking and query intents.
- **Multi-Intent Queries:** Some user inputs may involve multiple intents at once. For instance, a query like “Cancel my booking and send me the receipt” combines two different actions — cancellation, and sending the receipt

✓ ERROR ✓ WARNING ✓ INFO

Search agent issues

Agent Issues

1 1

There are no negative examples in the agent. Please create Default Fallback intent and fill it with examples.

There is no fallback intent in the agent.

Search intent issues



Intent Issues

4

app.current.notifications.open ✎

2

Parameter 'ad' has an empty value.

Multiple intents share training phrases which are too similar:

- Intent 'app.notifications.open': training phrase 'open notifications settings'
- Intent 'app.current.notifications.open': training phrase 'open notifications settings'

Handling Out-of-Scope Queries

Users may submit queries that fall outside the scope of the predefined intent categories. Handling out-of-scope queries is vital to prevent misleading or irrelevant responses:

- **Identifying Out-of-Scope Queries:** Developing mechanisms to recognize when a query does not align with known intent categories.
- **Providing Appropriate Responses:** Rather than giving a random or incorrect response, an agent should gracefully acknowledge that it cannot fulfil the request and offer assistance or redirection.



Dialogflow



Entities

Extracting useful information from user expressions

It shows what kind of information a word/phrase carries in a user input



JeroenGroet ▾ ⚙

en +

Intents +

Entities +

Knowledge [beta]

Fulfillment

Integrations

Training

Validation

History

Analytics

Prebuilt Agents

Small Talk

> Docs ↗

Trial
Free[Upgrade](#)Dialogflow CX ^[new]

tea

SAVE

[Try it now](#)☒ Define synonyms ⓘ ☐ Regexp entity ⓘ ☐ Allow automated expansion ☒ Fuzzy matching ⓘ

Separate synonyms by pressing the enter, tab or ; key.



English blend	Enter synonym	
Earl grey	Earl grey	
Green tea lemon	Green tea lemon	
Ginger lemongrass	Ginger lemongrass	
Sleepy time tea	Sleepy time tea	

[+ Add a row](#)

ⓘ Please use test console above to try a sentence.



JeroenGroet ▾ ⚙

en +

Intents +

Entities +

Knowledge ^[beta]

Fulfillment

Integrations

Training

Validation

History

Analytics

Prebuilt Agents

Small Talk

Docs

Trial
Free

[Upgrade](#)

Dialogflow CX ^[new]



Intents

CREATE INTENT



Search intents



Default Fallback Intent

Default Welcome Intent

Greeting

RequestDrink

Try it now



Please use test console above to try a sentence.



JeroenGroet ▾ ⚙

en +



Intents +



Entities +



Knowledge [beta]



Fulfillment



Integrations



Training



Validation



History



Analytics



Prebuilt Agents



Small Talk



Docs

Trial

Free

[Upgrade](#)

Dialogflow CX [new]

● RequestDrink

SAVE



Contexts ? ▾

Events ? ▾

Training phrases ?

Search training phrases



” Add user expression

” I'm craving for a ginger lemongrass!

” Do you have a sleepy time tea for me?

” Regular English blend for me

” Could you make me a Green tea lemon?

” I would like the Earl Grey, please!

Action and parameters ^

RequestDrink



REQUIRED ?	PARAMETER NAME ?	ENTITY ?	VALUE	IS LIST ?
☐	person	@sys.person	\$person	☐
☐	color	@sys.color	\$color	☐

Try it now



ⓘ Please use test console above to try a sentence.

JeroenGroet ▾ ⚙

en +

Intents +

Entities +

Knowledge [beta]

Fulfillment

Integrations

Training

Validation

History

Analytics

Prebuilt Agents

Small Talk

> Docs 📄

Trial
Free

Upgrade

Dialogflow CX [new]

Support 📄

RequestDrink

SAVE



Training phrases ?

Search training phrases



” Add user expression

” I'm craving for a ginger lemongrass!

” Do you have a sleepy time tea for me?

PARAMETER NAME

@tea:tea



RESOLVED VALUE

tea

Filter

sleepy time tea



” Regular English blend

@sys.time

” Could you make me

@sys.zip-code

” I would like the Earl Grey

@sys.street-address

@sys.unit-speed

Action and parameters

@sys.last-name

@sys.given-name

+ Create new

REQUIRED ?

PARAMETER NAME

RESOLVED VALUE

IS LIST ?

PROMPTS ?



tea

@tea

Stea



Define prompts...

+ New parameter

JeroenGroet ▾ ⚙

en +

Intents +

Entities +

Knowledge [beta]

⚡ Fulfillment

🔄 Integrations

🎓 Training

✅ Validation

🕒 History

📊 Analytics

📄 Prebuilt Agents

💬 Small Talk

> Docs 📄

Trial
Free Upgrade

Dialogflow CX [new]

🔗 Support 📄

RequestDrink

SAVE



Try it now



Contexts ? ▾

Events ? ▾

Training phrases ?

Search training phrases 🔍



” Add user expression

” I'm craving for a ginger lemongrass!

” Do you have a sleepy time tea for me?

” Regular English blend for me

” Could you make me a Green tea lemon?

” I would like the Earl Grey, please!

PARAMETER NAME	ENTITY	RESOLVED VALUE	
tea	@tea	Earl Grey	✕

Action and parameters ▴

RequestDrink

REQUIRED ?	PARAMETER NAME ?	ENTITY ?	VALUE	IS LIST ?	PROMPTS ?
☒	tea	@tea	Stea	☐	Define prompts...

Agent

USER SAYS

COPY CURL

I'd like an earl grey please

🗨️ DEFAULT RESPONSE ▾

Coming right up!

INTENT

RequestDrink

ACTION

RequestDrink

PARAMETER

VALUE

tea

earl grey

DIAGNOSTIC INFO

Get things up and running today!

Focus today on [getting your agent up and running \(Project Installation 2026\)](#):

- Clone the GitHub Classroom repository
- One team member creates and shares Dialogflow agent.
- Install the Social Interaction Cloud (version 2):

<https://socialrobotics.atlassian.net/wiki/spaces/PCA2/overview#Project-Installation>

- MARBEL plugin: *SET PREFERENCES AS INDICATED* + add Dialogflow agent.
- Launch to check you are ready!

Summary

This week: *Check out the [course schedule \(2026\)](#)!*

- Use **Canvas** “Discussion” → For ALL communication!
- Obligatory lab sessions !!